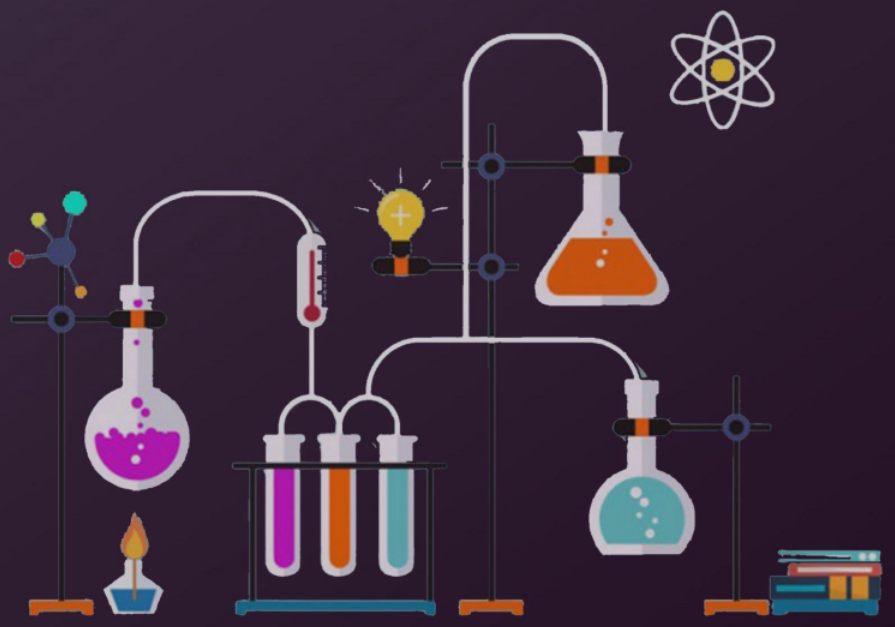
CHEMISTRY

Nomenclature



SHEET

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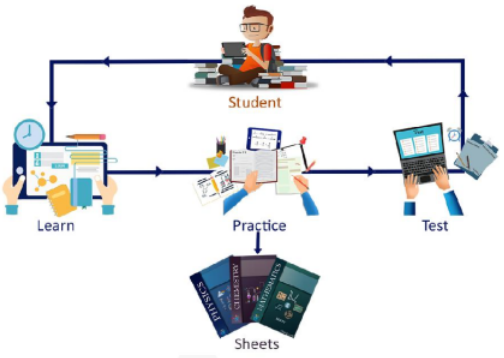
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 4
5.	Exercise 2	Page - 8
6.	Exercise 2A	Page - 12
7.	Answer Key	Page - 16

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Basics of Organic Chemistry	1	1-2, 13	1-3	17-18
2	Functional groups and degree of unsaturation	2-5	3, 6-7, 14	4-7	
3	Common Naming of Organic Compounds	6	10-11		
4	Rules of IUPAC Nomenclature	7-8			
5	Nomenclature of Saturated, Unsaturated and Complex Compounds		4	8-9	1-3, 8-11
6	Nomenclature of Cyclic Compounds	9-15		10	
7	Nomenclature of Mono Functional Group Containing Compounds	16-25	5, 15	11-19	4-6,19
8	Nomenclature of Poly Functional Group Containing Compounds	26-27	8-9	20-27	7, 12
9	Nomenclature of Bicyclo, Spiro and Aromatic Compounds	28	12	28-32	13-16
10	ACC	29		33	

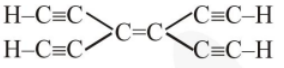
Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Basic of Organic Chemistry

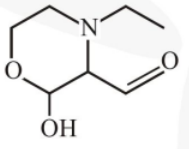
Q1. Total number of σ and π bond in given compound :



(A) 19σ, 9π (B) 9σ, 19π
(C) 8σ, 8π (D) 13σ, 9π

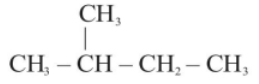
Functional Groups and Degree of Unsaturation

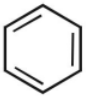
Q2. Which of the following functional group is not present in given compound :-



(A) Ketone (B) Aldehyde
(C) Ether (D) Amine

Q3. Which of the following compound has maximum DU?

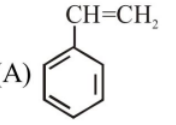
(A) 

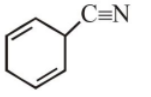
(B) 

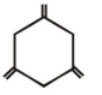
(C) 

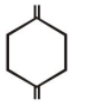
(D) 

Q4. Which of the following compounds has D.U. equal to benzene ?

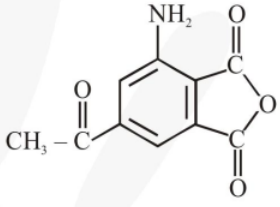
(A) 

(B) 

(C) 

(D) 

Q5. Which of the following functional group is absent in following molecule and what is the value of D.U.?



(A) Ether, 8 (B) Ketone, 7
(C) Amine, 8 (D) anhydride, 6

Common Naming of Organic Compounds

Q6. The **correct** statement for Iso-octane is –

(A) It has seven carbons in its principal carbon chain.
(B) It has five secondary carbons.
(C) It has one quarternary carbon and one tertiary carbon.
(D) It has ten primary hydrogens.

Rules of IUPAC Nomeclature

Q7. Which of the following is of highest priority if present in an compound ?

(A) $-SO_3H$ (B) –CHO
(C) –COOR (D) $-NH_2$

Q8. The priority of groups – OH, – COOH, – CHO & – OCH_3 attached to carbon is in order :

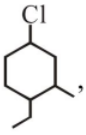
(A) – COOH > – CHO > – OH > – OCH_3

(B) – CHO > – COOH > – OCH_3 > – OH

(C) – OH > – OCH_3 > – CHO > – COOH

(D) – OCH_3 > – OH > – COOH > – CHO

Nomenclature of Cyclic Compounds

Q9. 

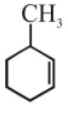
, IUPAC name of the compound is :-

(A) 5-chloro-2-ethyl-1-methyl-cyclohexane

(B) 1-chloro-4-ethyl-3-methyl-cyclohexane

(C) 1-ethyl-2-methyl-4-chloro cyclohexane

(D) 4-chloro-1-ethyl-2-methyl cyclohexane

Q10. IUPAC name of 

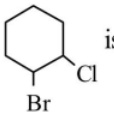
is -

(A) 3-methyl cyclohexene

(B) 1-methyl cyclohex-2-ene

(C) 6-methyl cyclohexene

(D) 1-methyl cyclohex-5-ene

Q11. The IUPAC name of the compound 

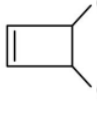
is -

(A) 1-Chloro-2-bromocylohexane

(B) 1, 2-Bromochlorocyclohexane

(C) 4-Bromo-3-chlorocyclohexane

(D) 1-Bromo-2-chlorocyclohexane

Q12. The IUPAC name of 

is -

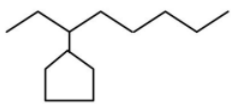
(A) 3-Methyl cyclo-1-buten-2-ol

(B) 4-Methyl cyclo-2-buten-1-ol

(C) 4-Methyl cyclo-1-buten-3-ol

(D) 2-Methyl cyclo-3-buten-1-ol

Q13. The IUPAC name of the given compound is-



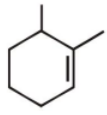
(A) Octylcyclopentane

(B) 3-Cyclopentyloctane

(C) Cyclopentaneoctane

(D) 6-Cyclopentyloctane

Q14. The IUPAC name of the structure is :-



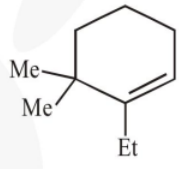
(A) 1,2-Dimethyl cyclohexane

(B) 1,6-Dimethyl cyclohexene

(C) 1,2-Dimethyl cyclohex-2-ene

(D) 2,3-Dimethyl cyclohexane

Q15. IUPAC name of the following compound :-



(A) 1-Ethyl–2, 2–dimethyl–6–cyclo hexene

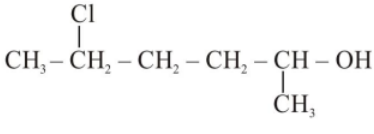
(B) 2-Ethyl–1, 1–dimethyl–2–cyclo hexene

(C) 1-Ethyl–6, 6–dimethyl–1–cyclo hexene

(D) 2-Ethyl–3, 3–dimethyl–1–cyclo hexene

Nomenclature of Mono Functional Group Containing Compounds

Q16. Correct IUPAC name of given compound

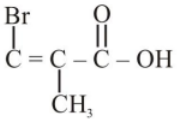


(A) 4–chloro–1–methylpentane–1–ol

(B) 5–chlorohexan–2–ol

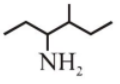
(C) 2–chlorohexan–5–ol

(D) 2–chlorohexan–2–ol

Q17. IUPAC name of 

(A) 3–Bromo–2–methylbutanoic acid
(B) 3–Bromo–2–methylbut–2–enoic acid
(C) 2–Bromo–3–methylbut–3–enoic acid
(D) 2–Bromo–3–methylbut–2–enoic acid

Q18. IUPAC name of the compound :

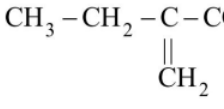


(A) 4-methylhexamine
(B) 3-methylhexamine
(C) 4-methylhex- 3-amine
(D) 4-ethylpent- 3-amine

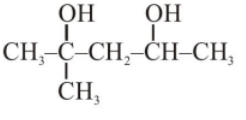
Q19. IUPAC name of $CH_3–CH_2–\underset{\diagdown O \diagup}{CH}–CH_2$ is :-

(A) 1,2-epoxy butane
(B) Ethyl methyl ether
(C) Keto pentanone
(D) None of these

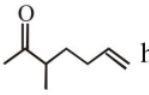
Q20. The correct IUPAC name of



(A) 2-Methyl butanoic acid
(B) 2-Ethyl prop-2-enoic acid
(C) 2-Carboxybutene
(D) None of the above

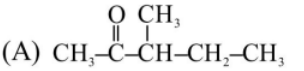
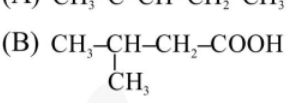
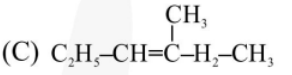
Q21. The IUPAC name of 

is :
(A) 4-methyl-2, 4, pentanediol
(B) 1, 1-dimethyl 1, 1, 3 butanediol
(C) 2-methyl-2, 4 pentanediol
(D) 1, 2, 3-trimethyl-1, 3 propanediol

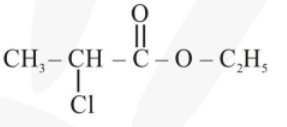
Q22.  has the IUPAC name :-

(A) 3-Methyl-6-heptene-2-one
(B) 5-Methyl-1-heptene-6-one
(C) 4-Allyl-3-methyl-2-butanone
(D) None of the above

Q23. 3–Methyl–2–pentanone is :-

(A) 
(B) 
(C) 
(D) $CH_3–CH_2–CH=CH_2$

Q24. Give IUPAC name of the given compound



(A) Methyl–2–chloropropanoate
(B) Chloro ethyl propanoate
(C) Ethyl–2–chloro propanoate
(D) Ethyl–2–chloro propane carboxylate

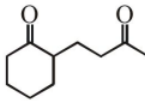
Q25. IUPAC name of $CH_3–\underset{CH_3}{CH}–CH_2–\underset{CN}{CH}–CH_3$:-

(A) 2-cyano-3-methyl, hexane
(B) 3-methyl-5-cyanohexane
(C) 2-4-dimethyl pentanenitrile
(D) 2-cyano-3-methylhexane

Nomenclature of Poly Functional Group Containing Compounds

Q26. The IUPAC name of $N\equiv C–CH_2–CH_2–OH$ is;

(A) 1-Hydroxy ethanenitrile
(B) 3-Hydroxy propanenitrile
(C) 2-Hydroxy ethyl cyanide
(D) 1-Hydroxy-2-cyanoethane

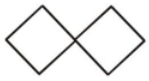
Q27. The IUPAC name of  is :-

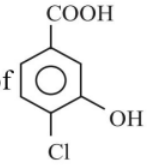
(A) 2–carboxyethyl cyclohexanone
(B) 3–(2–oxocyclo hexyl) propanoic acid
(C) Cyclo hexanone-2-propanoic acid
(D) 2–propanoic cyclo hexanone

Nomenclature of Bicyclo, Spiro and Aromatic Compounds

Q28. The structure of spiro [3, 3] heptane is :-

(A)  (B) 

(C)  (D) 

Q29. The IUPAC name of  is -

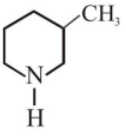
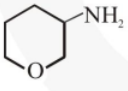
(A) 2 - Hydroxy - 1 - Chloro - 4 - Carboxy benzene
(B) 3 - Hydroxy - 4 - Chloro benzoic Acid
(C) 2 - Chloro - 4 - Carboxy phenol
(D) 4 - Chloro - 3 - Hydroxy benzene Carboxylic Acid

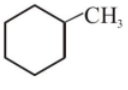
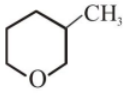
Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is/are hetrocyclic compound :-

(A)  (B) 

(C)  (D) 

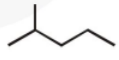
Q2. Which one are not secondary alcohol :-

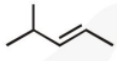
(A) 3-Methyl-1-butanol
(B) 2-Methyl-1-butanol
(C) 2,2-Dimethyl-1-propanol
(D) 3-Methyl-2-butanol

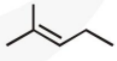
Q3. Which of the following is not correctly matched ?

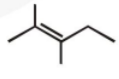
(A) ethanol - wood spirit
(B) acetic acid - ethanoic acid
(C) methanoic acid - formic acid
(D) Both (A) and (B)

Q4. The incorrect structure of 4-methyl-2-pentene is :-

(A) 

(B) 

(C) 

(D) 

Q5. Which of the following IUPAC name are correct :-

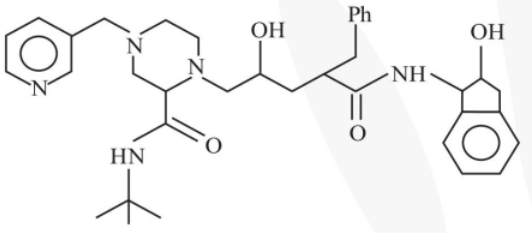
(A) Propan–1–ol
(B) But–1–ene
(C) Cyclohexanol
(D) Cyclohexanioc acid

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Q.No. 06 & 07

Crixivan, a drug produced by Merck and Co., is widely used in the fight against AIDS (acquired immune deficiency syndrome). The structure of crixivan is given below:



Q6. How many 2° alcohol groups are present in the above compound?

(A) Zero (B) 1

(C) 2 (D) 3

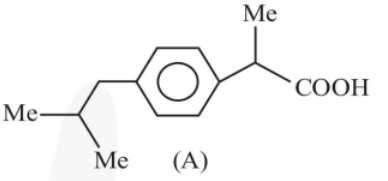
Q7. How many amide groups are present in the compound?

(A) Zero (B) 1

(C) 2 (D) 3

Paragraph for Q.No. 8 & 9

The analgesic drug ibuprofen (A) is chiral and exists in (+) and (–) forms. One enantiomer is physiologically active, while the other is inactive. The structure of ibuprofen is given below:



(A)

Q8. The principal functional group in (A) is:

(A) Phenyl (B) –COOH group

(B) Isopropyl (D) Both (1) and (2)

Q9. The IUPAC name of (A) is:

(A) 3 – (*p* – Isobutyl phenyl) propanoic acid

(B) 2 – (*p* – Isobutyl phenyl) propanoic acid

(C) 3 – (*p* – sec – Butyl phenyl) propanoic acid

(D) 2 – (*p* – sec – Butyl phenyl) propanoic acid

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q10.

Column - I (Common Name)	**Column - II (Structural formula)**
(A) Isooctane	(P) CH_2-CH_2 (Cl, Cl)
(B) Neopentane	(Q) $CH_3-C(CH_3)(CH_3)-CH_2-CH(CH_3)-CH_3$

(C) Ethylidene chloride (Geminal dihalide)

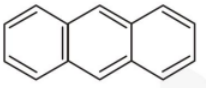
(R) $CH_3 - CH_2 - CH_2 - CH_2 - CH_2 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_3$

(D) Ethylene Dichloride (Vicinal dihalide)

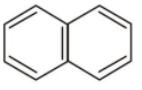
(S) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle CH_3}{\overset{|}{C}}}} - CH_3$

(T) $CH_3 - CH\begin{matrix} Cl \\ Cl \end{matrix}$

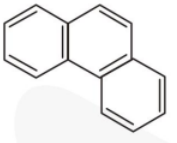
Q11. **Column - I** **Column - II**

(A) 

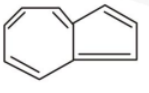
(P) Phenanthrene

(B) 

(Q) Anthracene

(C) 

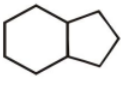
(R) Azulene

(D) 

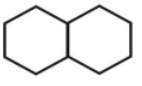
(S) Napthalene

Q12. Match column I with column II and select the correct answer from the given codes :

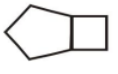
Column-I (compounds) **Column-II (number of carbons in the bridges)**

(A) 

(P) [3.2.1]

(B) 

(Q) [4.3.0]

(C) 

(R) [4.4.0]

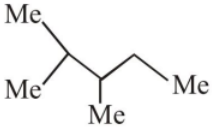
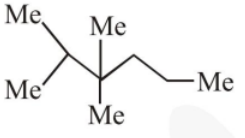
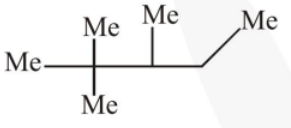
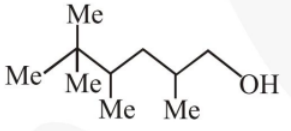
(D) 

(S) [3.2.0]

MATCHING LIST TYPE

Each question has TWO (02) ***matching lists: LIST I and LIST II****. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q13.

List - I (Compound)	**List - II (Nature of H-atoms)**
(P) 	(1) 15(1°H), 4(2°H), 1(3°H)
(Q) 	(2) 17(1°H), 2(2°H), 2(3°H)
(R) 	(3) 12(1°H), 2(2°H), 2(3°H)
(S) 	(4) 15(1°H), 2(2°H), 1(3°H)

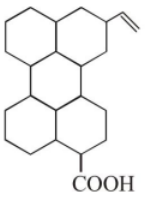
Code :

	P	Q	R	S
(A)	3	1	4	2
(B)	3	2	1	4
(C)	1	2	3	4
(D)	3	1	2	4

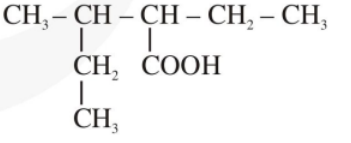
NUMERICAL TYPE

The answer to each question is a ***NUMERICAL VALUE****. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q14. Double bond equivalent (DBE) of the following compound.



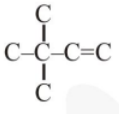
Q15. Number of carbon atoms in the principal carbon chain of the given compound in its IUPAC name will be.



SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Which of the following statement is correct about the given compound :-



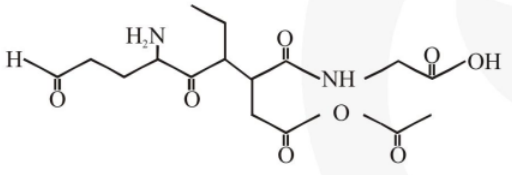
(A) Given compound has '4' primary carbon and '1' secondary hydrogen

(B) Given compound has 11 primary hydrogen but no tertiary carbon

(C) Given compound has '1' quarternary carbon and two 'sp^2' hybridized carbon

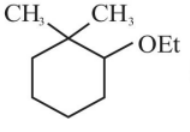
(D) All of these

Q2. The number of functional groups present in the following compound is :-



(A) 5 (B) 7

(C) 6 (D) 8

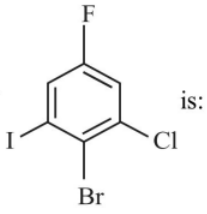
Q3. IUPAC name of  is:

(A) 1 – Ethoxy – 2,2 – dimethylcyclohexane

(B) 2 – ethoxy – 1,1 – dimethyl cyclohexane

(C) 1, 1 – Dimethyl – 2 – ethoxycyclohexane

(D) 2 – methyl – 1, 1 – ethoxy cyclohexane

Q4. The IUPAC name of  is:

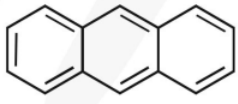
(A) 1 – Bromo – 2 – chloro – 3 – fluoro – 6 – iodo benzene

(B) 2 – Bromo – 1– chloro – 5 – fluoro – 3 – iodo benzene

(C) 4 – Bromo – 2 – chloro – 5 – iodo – 1 – fluoro benzene

(D) 2 – Bromo – 3 – chloro – 1 – iodo – 5 – fluoro benzene

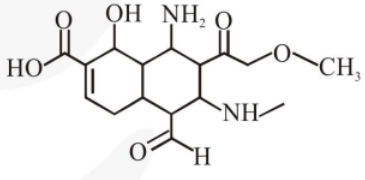
Q5. Number of σ, π bond and D.U. in anthracene:-



(A) 16, 7, 10 (B) 16, 5, 7

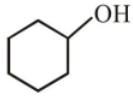
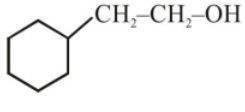
(C) 30, 7, 10 (D) 26, 7, 10

Q6. How many functional groups are present in following compound?



(A) 7 (B) 6

(C) 8 (D) 9

Q7. (a)  & (b) 

True statement for the above compounds is:-

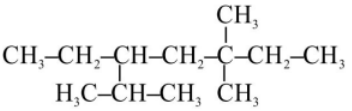
(A) (a) is phenol while (b) is alcohol

(B) Both (a) & (b) are primary alcohol

(C) (a) is primary & (b) is secondary alcohol

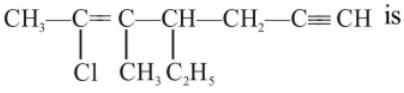
(D) (a) is secondary & (b) is primary alcohol

Q8. The IUPAC name of the structure is :-



(A) 3–Isopropyl–5,5–dimethyl heptane
(B) 5–Ethyl–3,3,6–trimethyl heptane
(C) 3,3–Dimethyl–5–isopropyl heptane
(D) 3–Ethyl–2,5,5–trimethyl heptane

Q9. The IUPAC name of

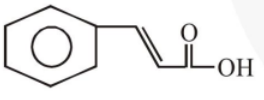
 is :-

(A) 6-chloro-4-ethyl-5-methyl-hept-5-en-1-yne
(B) 6-chloro-4-ethyl-5-methyl-hept-1-yn-5-ene
(C) 2-chloro-4-ethyl-3-methyl-hept-2-en-6-yne
(D) 2-chloro-4-ethyl-3-methyl-hept-6-yn-2-ene

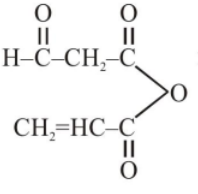
Q10.  IUPAC name is :-

(A) 6-methyl-3,3-diethyl cyclohexene
(B) 3-ethyl-6,6-dimethyl cyclohexene
(C) 5-ethyl-3,3-dimethyl cyclohexene
(D) 4-ethyl-6,6-dimethyl cyclohex-2-ene

Q11. The correct IUPAC name of compound is:-

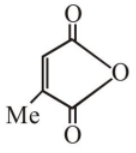


(A) cinnamic acid
(B) 2-methyl-prop-2-enoic acid
(C) 2-phenyl-prop-2-enoic acid
(D) 3-phenyl-prop-2-enoic acid

Q12. The IUPAC name of  is :-

(A) 2-Formylethanoic propanoic Anhydride
(B) 2-Oxopropanoic prop-2-enoic Anhydride
(C) Prop-2-enoic-2-formyl propanoic Anhydride
(D) 3-Oxopropanoic prop-2-enoic Anhydride

Q13. The correct IUPAC name of the compound:



(A) 2-Methylbut-2-ene dioic anhydride
(B) 3-Methylbut-2-ene dioic anhydride
(C) 2-Methyl-1,4-diketobut-2-ene epoxy
(D) 2-Methyl-cyclo pentanoxy-1,4-dione

Q14. Which of the following IUPAC name is wrong :-

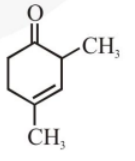
(A) $CH_3CH_2CH_2COOCH_2CH_3$ ethylbutanoate

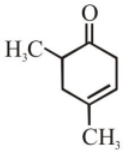
(B)  3-methylbutanal

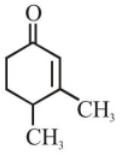
(C)  2-methyl-3-butanol

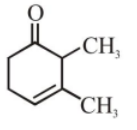
(D)  2-methyl-3-petanone

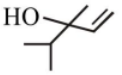
Q15. 2,4-Dimethyl-3-cyclohexenone can be written as :-

(A) 

(B) 

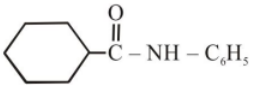
(C) 

(D) 

Q16.  has the IUPAC name :-

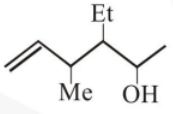
(A) 3,4–Dimethyl–1–penten–3–ol
(B) Isopropyl–3–methyl vinyl carbinol
(C) 2,3–Dimethyl–4–penten–3–ol
(D) None of the above

Q17. The IUPAC name of :-



(A) N – cyclohexyl benzamide
(B) N–phenyl–N–cyclohexylmethanamide
(C) N – phenylcyclohexane carboxamide
(D) N – cyclohexyl – N – phenylmethylamide

Q18. The IUPAC name of following compound is -

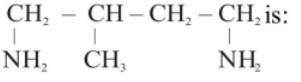


(A) 3-Ethyl-4-methylhex-5-en-2-ol
(B) 4-Ethyl-3-methylhex-1-en-5-ol
(C) 3-Methyl-4-ethylhex-1-5-ol
(D) 4-methyl-3-ethylhex-5-en-2-ol

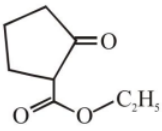
Q19. Which is the correct structure of the compound 3-hexyne-1-oic acid :-

(A) $CH_3–CH_2–CH_2–C\equiv C–COOH$
(B) $CH_3–CH_2–C\equiv C–CH_2–COOH$
(C) $CH_3–C\equiv C–CH_2–CH_2–COOH$
(D) $CH_3–CH_2–CH=CH–CH_2–COOH$

Q20. The IUPAC name of

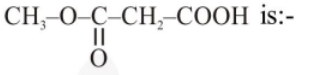


(A) 2 – Methylbutane – 1, 4 – diamine
(B) 3 – Methylbutane – 1, 4 – diamine
(C) 3 – (Aminomethyl) butanamine
(D) 2 – (Aminomethyl) butan – 4 – amine

Q21. The compound  has been named in IUPAC as :-

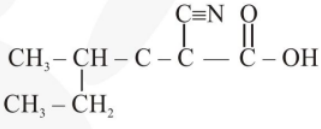
(A) ethyl-2-oxocyclopentanecarboxylate
(B) 1-oxo-2-carbethoxy cyclopentane
(C) Carbethoxy cyclo pentanone
(D) None of these

Q22. The IUPAC name of



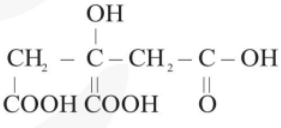
(A) 2-Acetoxyethanoic acid
(B) 2-Methoxycarbonylethanoic acid
(C) 3-Methoxyformylethanoic acid
(D) 2-Methoxyformylacetic acid

Q23. The number of carbon atom in parent carbon chain in given compound is ?



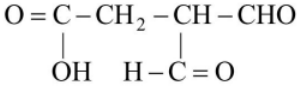
(A) 6 (B) 5
(C) 3 (D) 7

Q24. The correct systematic IUPAC name of the given compound is :

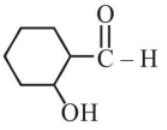


(A) 3-carboxy-3-hydroxybutane dioic acid
(B) 2-Hydroxy propane-1,2,3-tri carboxylic acid
(C) 3-Hydroxy butane dioic acid
(D) 2-Bis(carboxymethyl)-2-hydroxy ethanoic acid

Q25. The correct IUPAC name of the following compound is:

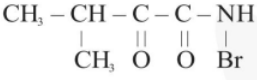


(A) 3,3-Diformylpropanoic acid
(B) 3-Formyl-4-oxo-butanoic acid
(C) 3,3-Dioxo propanoic acid
(D) 3,3-Dicarbaldehyde propanoic acid

Q26. IUPAC name of 

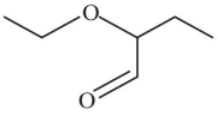
(A) 2–Hydroxy cyclohexane carbaldehyde
(B) 2–Formylcyclohexanol
(C) 1–Formyl–2–hydroxycyclohexane
(D) 2–Hydroxycyclohexanol

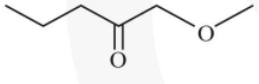
Q27. The IUPAC name of

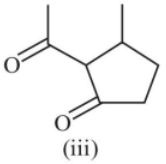


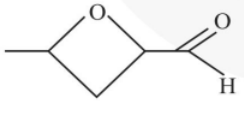
(A) (*N* – Bromo) – 2 – keto – 3 – methylbutanamide
(B) (*N* – Bromo) – 2 – keto – 4 – methylbutanamide
(C) (*N* – Bromo) – 1, 2 – diketo – 3 – methylbutanamine carboxamide
(D) (*N* – Bromo) – 1 – keto – 2 – methylpropane

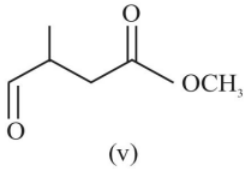
Q28. Many organic compounds contain more than one functional group. In which of the following is both aldehyde and ether are present?


(i)

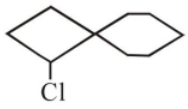

(ii)


(iii)


(iv)

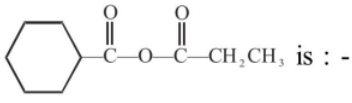

(v)

(A) (i) and (iii)
(B) (i) and (iv)
(C) (ii) and (v)
(D) (iii) and (iv)

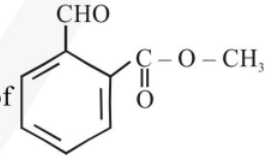
Q29.  has the IUPAC name -

(A) 3-chloro spiro (3,5) nonane
(B) 1-chloro spiro (3,5) nonane
(C) 1-chloro spiro (5,3) nonane
(D) 3-chloro spiro (5, 3) nonane

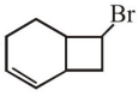
Q30. The IUPAC name of

 is : -

(A) cyclohexanoic propanoic anhydride
(B) cyclohexane carboxylic propanoic anhydride
(C) cyclohexyl ethyl anhydride
(D) benzoic propanoic anhydride

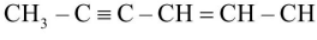
Q31. I.U.P.A.C name of 

(A) O – Methoxy carbonyl benzaldehyde
(B) 2 – Formyl benzyl methanoate
(C) Methyl – 2-formyl benzoate
(D) Methyl salicylate

Q32. The IUPAC name of  :-

(A) 7-Bromobicyclo [4, 2, 0]-2-octene
(B) 2-Bromobicyclo [4, 2, 0]-5-octene
(C) 8-Bromobicyclo [2, 4, 0]-4-octene
(D) 6-Bromobicyclo [4, 2, 0]-1-octene

Q33. Find out the incorrect statement about given molecule :-



(A) IUPAC name is hepta-1,3-dien-5-yne
(B) It has 4 sp^2 hybrid carbons, 2sp hybrid carbons and 1sp^3 hybrid carbon
(C) All the carbons in given molecule are in same plane
(D) All the carbons in given molecule are not in same plane

MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following statements is/are wrong?

(A) Acetic acid is the systematic name of vinegar.

(B) $Me-\overset{\overset{O}{\|}}{C}-OH$ is an unsaturated hydrocarbon.

(C) Prefixes like n–, iso, sec–, tert, neo–, etc., are used in IUPAC system.

(D) The systematic names of acids are formed by dropping –e of the name of parent alkane and adding –oic acid.

Q2. Which of the following represent incorrect IUPAC name :

(A) 2-Ethyl Pentane

(B) 3-Methyl Pentane-1-one

(C) 3-Ethyl Pentane

(D) 2-(1-Methyl ethyl) Butane

Q3. Name of some compounds are given. Which one is correct in IUPAC system :-

(A) $CH_3-CH_2CH_2\underset{\underset{CH_2CH_3}{|}}{CH}-\overset{\overset{CH_3}{|}}{CH}-CH_2CH_3$;

3-Methyl-4-ethyl heptane

(B) $CH_3\underset{\underset{OH}{|}}{CH}-\underset{\underset{CH_3}{|}}{CH}CH_3$;

3-Methyl-2-butanol

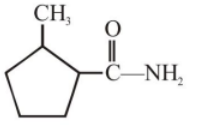
(C) $CH_3CH_2-\underset{\underset{CH_2CH_3}{\|}}{C}-\underset{\underset{CH_3}{|}}{CH}CH_3$;

2-Ethyl-3-methyl-but-1-ene

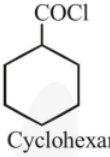
(D) $CH_3-C\equiv C-CH(CH_3)_2$;

4-Methyl-2-pentyne

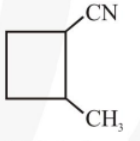
Q4. For the which of the following compounds, the IUPAC name are correct ?

(A) 

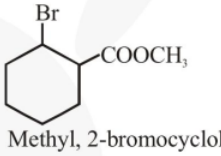
2-methylcyclopentane carboxamide

(B) 

Cyclohexaneoyl chloride

(C) 

2-Methyl cyclobutane carbonitrile

(D) 

Methyl, 2-bromocyclohexane carboxylate

Q5. The incorrect name of 2-chloro butan-3-ol are :-

(A) 3-chloro-2-hydroxy butane

(B) 3-chloro-2-butanol

(C) 3-hydroxy-2-chloro-butane

(D) 2-chloro-3-hydroxy butane

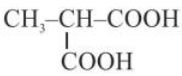
Q6. Which of the following are not correct according to the IUPAC nomenclature

(A) 1, 2, 3-trihydroxy propane $\begin{array}{l}CH_2-OH\\|\\CH_2-OH\\|\\CH_2-OH\end{array}$

(B) 2-Ethyl–3–methyl butane,

$CH_3-\underset{\underset{Et}{|}}{CH}-\underset{\underset{Me}{|}}{CH}-CH_3$

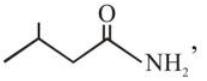
(C) 2–Carboxypropanoic acid,



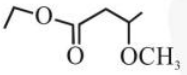
(D) 6–Methylbicyclo [3.2.1]oct–2–ene



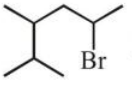
Q7. Which are correctly named according to IUPAC nomenclature :-

(A)  ,

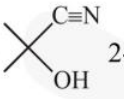
3-Methylbutanamide

(B) 

Ethyl-3-methoxy butanoate

(C)  2-Bromo-4,

5-dimethyl hexane

(D)  2-Hydroxy-2-mehthylpropane cyanide

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 8 to 11

Branched- chain alkanes are named according to the following rules.

(1) Longest chain Rule -

Locate the longest continuous chain of carbon atoms. This chain determines the parent name of the alkane.

(2) Lowest set of locants - The longest continuous chain are numbered by arabic numerals 1, 2, 3, 4,.... from one end of chain to the other, in such a manner that carbon atom carrying first substituent gets the lowest number.

(3) Name of the branched chain alkane - The substituent name and the parent alkane are joined in one word and there is a hyphen between the number and the substituent name.

(4) Alphabetical order of the side chains - When two or more substituents are present, give each substituent a number corresponding to its position on the longest chain, the substituent group should be listed alphabetically.

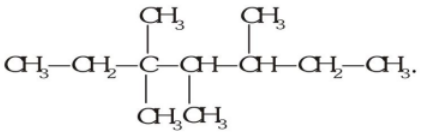
(5) Numbering of different alkyl groups at equivalent positions- If two different alkyl groups are present at equivalent positions the numbering of the parent chain is done in such a way that alkyl group which comes first in the alphabetical order gets the lower number.

(6) Naming of same alkyl groups at different positions - When two or more substituents are identical, indicate this by the use of prefixes di, tri, tetra and soon. Commas are used to separate numbers from each other.

(7) Rule for larger number of substituents - If a compound has two or more chains of the same length, the parent hydrocarbon is the chain with the greater number of substituents.

(8) Numbering the complex substituent - Name such as iso-propyl, sec-butyl and tert-butyl are acceptable substituent name in the IUPAC system of nomenclature but systematic name are preferable. Systematic substituent names are obtained by numbering the substituent starting at the carbon that is attached to the parent hydrocarbon. This means that the carbon that is attached to the parent hydrocarbon is always the number –1carbon of the substituent.

Q8. In following compound

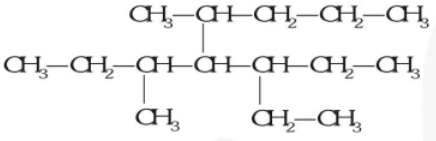


The correct lowest set of locant is :-

(A) 3, 3, 4, 5 (B) 3, 4, 5, 5

(C) 4, 5, 3, 3 (D) 5, 5, 4, 3

Q9. The IUPAC name of the compound is ?



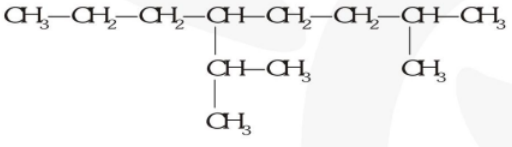
(A) 5-Ethyl-3-Methyl-4-(1-Methylpropyl) octane

(B) 4-Ethyl-6-Methyl-5-(1-Methylpropyl) octane

(C) 3-Ethyl-5-Methyl-4-(1-Methylpropyl) octane

(D) 4-Sec-butyl-5-Ethyl-5-Methylheptane

Q10. The correct IUPAC name of the following compound is -



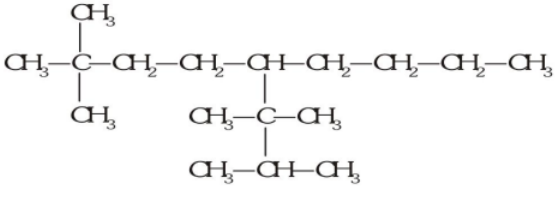
(A) 2-Methyl-5-(1-Methylethyl)octane

(B) 2, 6-Dimethyl-5-propylheptane

(C) 5-propyl-2-Methyloctane

(D) 4-(1-Methylethyl)-7-methyloctane

Q11. The correct IUPAC name of following compound is -



(A) 2, 3, 3, 7, 7 Pentamethyl-4-butyloctane

(B) 4-Butyl-2, 3, 3, 7, 7 pentamethylnonane

(C) 2, 2-Dimethyl-5-(1', 1', 2'-trimethylpropyl)nonane

(D) 5-(1', 1', 2'-trimethylpropyl)-2, 2-dimethylnonane

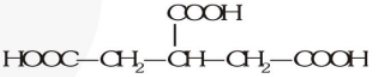
Paragraph for Question No. 12

If the organic compond contains more than two similar terminal groups and all of them are directly attached to the principal chain, then none of them forms a part of the principal chain. Special suffixes are used to name these :-

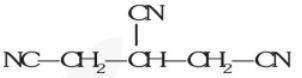
Functional group	Suffix
—$CONH_2$	Carboxamide
—CN	Carbonitrile
—CHO	Carbaldehyde
—COOH	Carboxylic acid

Carbon atoms of these terminal groups are not counted in the principal chain. If any one of these terminal groups is not directly attached to the parent chain and forms the part of side chain, then the longest chain is selected containing two such similar groups at its two ends. The groups present in the side chain are treated as substituents and are indicated by suitable prefixes.

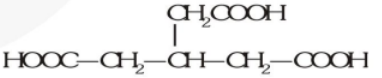
Q12. Indicate whether the following IUPAC names are true (a) or false (b)

1. 

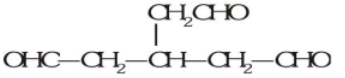
Propane-1,2,3-tricarboxylic acid

2. 

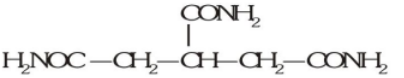
3-cyanopentane-1,5-dinitrile

3. 

3-(carboxymethyl)-1,5-dioic acid

4. 

3-(formylmethyl)pentane-1,5-dial

5. 

Propane-1,2,3-tricarboxamide

Paragraph for Question No. 13 to 15

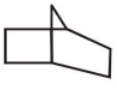
In addition to the standard ring systems (such as cyclohexane), cyclic compounds can also be bicyclic, tricyclic, etc. or they can be spirocyclic, bicyclic or bridge head carbons. The point of attachment of two rings are called bridge head atoms.

The formal names of bicyclic and related ring systems are based on

(a) Total numbe of atoms in the molecule.

(b) The number of atoms in each bridge connecting the bridge head atoms. These numbers are written in square bracket in decreasing order.

Spirocyclic compounds have two fused rings, but only one bridge head atom. Spirocyclic compounds are named like bicyclic compounds, but have the prefix spirocyclo. Answer the following question :

Q13. 

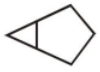
what is the IUPAC name of the above compound ?

(A) cyclo [1.2.2] heptane

(B) Bicyclo [1.2.2] heptane

(C) Bicyclo [2.2.1] heptane

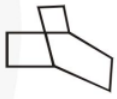
(D) cyclo [2.2.1] heptane

Q14.  The number of atoms in each bridge are :

(A) [3.2.1] (B) [3.1.0]

(C) [1.3.0] (D) [2.1.0]

Q15. Select the correct statements about the following compounds :



(A) It is a tricyclic compound

(B) It is bicyclo compound

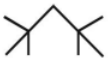
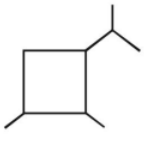
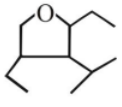
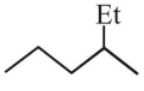
(C) It is spiro compound

(D) Its IUPAC name is bicyclo [2.2.2] hexane

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

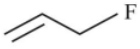
Q16.

Column-I	**Column-II**
(A) 	(P) Even number of 1° carbon, atom
(B) 	(Q) Odd number of 2° carbon atom
(C) 	(R) Homocyclic compound
(D) 	(S) Principal carbon chain is having odd number of carbon atoms

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q17. How many 1° carbon atom will be present in a simplest hydrocarbon having two 3° and one 2° carbon atom?

Q18.  When IUPAC name of following compound is given, then double bond and substituent gets respectively (x and y) number so the sum of $(x+y)$ will be:

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10
D	A	B	C	A	C	A	A	D	A
11	12	13	14	15	16	17	18	19	20
D	B	B	B	C	B	B	C	A	B
21	22	23	24	25	26	27	28	29	
C	A	A	C	C	B	B	D	D	

Ex - 1A

1. A,B,D **2.** A,B,C **3.** A **4.** A,C,D **5.** A,B,C

6. C **7.** C **8.** B **9.** B

10. A (Q), B (S), C (T), D (P)

11. A (Q), B (S), C (P), D (R)

12. A (Q), B (R), C (S), D (P)

13. A **14.** 7 **15.** 5

Ex-2

1	2	3	4	5	6	7	8	9	10
D	C	B	B	D	C	D	D	A	C
11	12	13	14	15	16	17	18	19	20
D	D	A	C	A	A	C	A	B	A
21	22	23	24	25	26	27	28	29	30
A	B	A	B	B	A	A	B	B	B
31	32	33							
C	A	D							

Ex - 2A

1. A,B,C **2.** A,B,D **3.** B,C,D **4.** A,C,D **5.** A,C,D

6. A,B,C **7.** A,B,C **8.** A **9.** C **10.** C

11. C **12.** 1(T), 2(F), 3(F), 4(T), 5(T)

13. C **14.** B **15.** B

16. (A)→P, Q, S ; (B)→P, Q, R ; (C)→Q, S ; (D)→Q

17. 4 **18.** 4

-: Important Notes & Formulas :-